

B.Tech III Year II Semester (R20) Supplementary Examinations January 2024

ADVANCED COMPUTER ARCHITECTURE

(Computer Science & Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Define System Software. | 2M |
| (b) What is meant by Assembler? | 2M |
| (c) What is meant by loader? | 2M |
| (d) Define linker. | 2M |
| (e) Give suitable examples for Binary Addition and Subtraction. | 2M |
| (f) Write the result after dividing 1001010_2 by 1000_2 . | 2M |
| (g) Define Clocking Methodology. | 2M |
| (h) What is the purpose of Program Counter (PC)? | 2M |
| (i) What is meant by write-through? | 2M |
| (j) Define rotational write-back. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain Technologies for Building Processors and Memory. 10M
- OR**
- 3 (a) If computer A runs a program in 10 seconds and computer B runs the same program in 15 seconds, how much faster is A than B? 5M
- (b) Write briefly about The Power Wall. 5M
- 4 Write about Instructions for Making Decisions in computer with suitable examples. 10M
- OR**
- 5 Describes in detail the four steps in transforming a C program in a file on disk into a program running on a computer with diagrams. 10M
- 6 (a) Add the numbers 6_{ten} to 7_{ten} in binary and then subtracting the numbers 6_{ten} from 7_{ten} in binary. 5M
- (b) Add the numbers 0.5_{ten} and 0.4375_{ten} in binary using the algorithm. 5M
- OR**
- 7 (a) Illustrate the problems in floating-point addition: $9.999_{ten} \times 10^1 + 1.610_{ten} \times 10^{-1}$. 5M
- (b) Explain how to Converting Binary to Decimal Floating Point with suitable example. 5M
- 8 Write in detail about an Overview of Pipelining. 10M
- OR**
- 9 Write summary on Data Hazards for Forwarding versus Stalling. 10M
- 10 Explain in Brief about Memory Technologies. 10M
- OR**
- 11 Write briefly about Parallelism and Memory Hierarchies. 10M

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

ADVANCED COMPUTER ARCHITECTURE

(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) Define Moore's Law. | 2M |
| (b) What is the role of benchmarks in measuring performance of computer systems? | 2M |
| (c) Give the addressing modes in MIPS. | 2M |
| (d) Interpret the instruction set Architecture. | 2M |
| (e) Analyze overflow conditions for addition and subtraction. | 2M |
| (f) State sub-word parallelism. | 2M |
| (g) Define hazard. Give an example for data hazard. | 2M |
| (h) What is meant by Pipelining? | 2M |
| (i) Draw the structure of memory hierarchy. | 2M |
| (j) What do you mean by cache coherence? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|-----|
| 2 | Explain the Eight Great Ideas in Computer Architecture in brief. | 10M |
| OR | | |
| 3 | Discuss the Switch from Uniprocessors to Multiprocessors. | 10M |
| 4 | Discuss the logical operations and control operations of computer. | 10M |
| OR | | |
| 5 | Explain the various techniques to represent instructions in a computer system. | 10M |
| 6 | (a) Explain the concept of Arithmetic operation with examples. | 5M |
| | (b) Write notes on ALU design. | 5M |
| OR | | |
| 7 | (a) Discuss the multiplication algorithm in detail with diagram. | 5M |
| | (b) Express the steps to multiply 2*3. | 5M |
| 8 | Design and develop an instruction pipeline working under various situations of pipeline stall. | 10M |
| OR | | |
| 9 | (a) Summarize control implementation scheme. | 5M |
| | (b) Distinguish the data and control path methods in pipelining. | 5M |
| 10 | Discuss virtual memory address translation in detail with necessary diagram. | 10M |
| OR | | |
| 11 | (a) Analyze the given problem: A byte addressable computer has a small data cache capable of holding. Eight 32-bit words. Each cache block contains 132-bit word. When a given program is executed, the processor reads data from the following sequence of hex addresses – 200, 204, 208, 20C, 2F4, 2F0, 200,204,218, 21C, 24C, 2F4. The pattern is repeated four times. Assuming that the cache is initially empty, show the contents of the cache at the end of each pass, and compute the hit rate for a direct mapped cache. | 5M |
| | (b) What are the methods used to measure and improve the performance of the cache? | 5M |

B.Tech III Year II Semester (R20) Regular Examinations August 2023

ADVANCED COMPUTER ARCHITECTURE

(Computer Science & Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Define computer hardware. | 2M |
| (b) Write the formula for finding performance & Relative Performance of computer. | 2M |
| (c) What is meant by data transfer instructions? | 2M |
| (d) Give Examples for least significant bit and most significant bit. | 2M |
| (e) Give suitable examples for Binary Addition and Subtraction. | 2M |
| (f) Write the result after Multiplying 1000_2 by 1001_2 . | 2M |
| (g) Define clocking methodology. | 2M |
| (h) What is the purpose of program counter (PC)? | 2M |
| (i) What is meant by Hit rate and Miss rate? | 2M |
| (j) Define rotational latency. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Write about Eight Great Ideas in Computer Architecture. 10M
- OR**
- 3 (a) What is meant by Throughput and Response Time? 5M
(b) Write briefly about CPU Performance and Its Factors. 5M
- 4 Explain in detail about Operands of the Computer Hardware with suitable examples where ever possible. 10M
- OR**
- 5 How do you Representing Instructions in the Computer give suitable examples? 10M
- 6 (a) Illustrate the Addition and subtraction of numbers 6_{ten} to 7_{ten} in binary. 5M
(b) Using a 4-bit version of the algorithm divide 7_{ten} by 2_{ten} , or $0000\ 0111_{two}$ by 0010_{two} . 5M
- OR**
- 7 (a) Explain Signed Division with suitable example. 5M
(b) Explain Floating-Point Representation and Show the IEEE 754 binary representation of the number 0.75_{ten} in single and double precision. 5M
- 8 Write briefly about Basic MIPS Implementation. 10M
- OR**
- 9 Explain briefly about Building a Datapath. 10M
- 10 Write about the Basics of Caches in detail. 10M
- OR**
- 11 Explain Virtual Machines in detail. 10M
